



Air University
Semester Examinations: Spring 2026

Student ID: _____

Subjective Part
(To be solved on Answer Books only)

Subject: Privacy and Usable Security
Class: BSCYS
Section(s): A
Course Code: CY-455

Time Allowed: 180 Minutes
Max Marks: 100
FM's Name: Dr. Sheikh Qaisar Ayyub
FM's Signature:

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.
- Tables / calculators are allowed / not allowed.

Q. No. 1 (CLO 3)		Marks:
		40
A	A user installs a popular mobile fitness application that has millions of downloads and strong ratings. During installation, a privacy policy appears with multiple sections and scrolling pages. The user briefly looks at the first few lines, sees technical and legal terms, and immediately clicks "Accept" to continue because they are in a hurry to start tracking their workout. After a few days: • The app starts showing highly targeted ads • The user receives promotional emails from unknown companies • The app tracks location even when not in use The user says: "I thought this was just a simple fitness app. I didn't expect all this." Answer Following • Why do users often ignore or quickly accept privacy policies without reading them carefully? • What behavioral or psychological factors influence this decision? • Identify the privacy risks that emerged after the user accepted the policy. • Explain how "consent fatigue" or "information overload" contributes to such situations. • Suggest two ways to improve user understanding of privacy policies in mobile applications.	25
B	• A banking web application is deployed for online transactions and accessed by thousands of users. After deployment, the system faces several issues such as brute-force login attempts, suspicious user activity, possible SQL injection vulnerabilities, and delayed security updates. The application remains live due to business requirements, but security concerns continue to increase. Answer Following • Identify the security risks present in the deployed system. • Explain how SQL injection testing, authentication bypass testing, and privilege escalation testing are relevant to this system.	15

	Suggest security measures that should be applied after deployment to protect the software.	
	Q. No. 2 (CLO 2)	Marks: 40
A	A university learning portal is designed for easy use, allowing students to quickly log in, submit assignments, and check results. To improve usability, it includes features like "remember me" login, simple passwords, long-lived sessions, and minimal security warnings. After deployment, users report unauthorized access and account misuse. Answer Following - Identify the usability-driven vulnerabilities in the system. - Explain how usability features can weaken security. - Suggest ways to balance usability and security in such systems.	15
B	A software company introduces an employee monitoring system to detect insider threats. It tracks login activity, messages, files, and browsing history. After deployment: - Employees are not clearly informed - Personal messages are also monitored - Stored data has weak security - A supervisor misuses data to view private chats - Employees feel stressed and constantly watched Answer Following - Identify any four privacy challenges in the scenario - Which issue can lead to loss of employee trust? - What type of problem is monitoring personal messages? - Which issue relates to poor data protection? - Suggest two improvements to protect employee privacy	25
	Q. No. 3 (CLO 3)	20
	Discuss the complete life cycle of a Machine Learning project from a privacy perspective. Your answer should explain how privacy concerns can arise during each phase of the ML life cycle, including: - Data - Features - Model - Prediction - Evaluation	10
	- Explain the System Usability Scale (SUS) in detail. How is the SUS score calculated? - Describe the Cognitive Walkthrough method. Explain how it helps identify usability problems in a system.	10